

MATLAB PROGRAM REPORT

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1 Introduction

Problem Statement 1 — Image Processing & Matrices

Title: Image Contrast Enhancement Using Matrix Operations

Write a MATLAB script that reads a grayscale image, converts it into a matrix, and applies the following operations:

1. Increase the contrast using linear stretching:

$$I_{new} = \frac{(I - \min(I))}{(\max(I) - \min(I))} \times 255$$

2. Display the original and enhanced images in a 2-subplot figure.
3. Use loops to manually compute the minimum and maximum pixel values without using `min()` or `max()`.
4. Save the enhanced image in a new file using MATLAB file-handling functions.

```
% Problem 1 Solution
img = imread('image.jpg');
img = rgb2gray(img);
imwrite(img, "input image.jpg");
[m,n] = size(img);

% Compute min/max manually
mn = 255; mx = 0;
for i = 1:m
    for j = 1:n
        if img(i,j) < mn
            mn = img(i,j);
        end
        if img(i,j) > mx
```

```

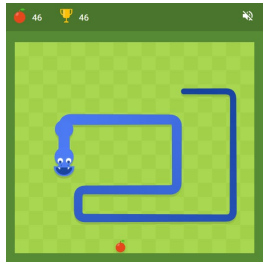
        mx = img(i,j);
    end
end
end

enh = double(img - mn) / double(mx - mn) * 255;
enh = uint8(enh);

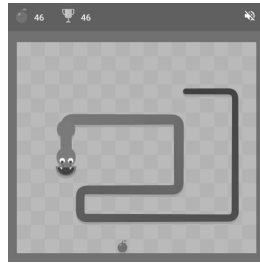
subplot(1,2,1), imshow(img), title('Original');
subplot(1,2,2), imshow(enh), title('Enhanced');

imwrite(enh, 'enhanced_output.png');

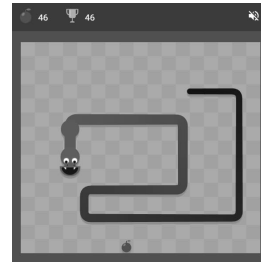
```



(a) Original Image



(b) Greyscale Image



(c) Enhanced Image

Figure 1: Input and Output of Problem 1